

LECTURE-5 GDA & Naive Bayes - Stanford

- Generative Learning Algorithm
 - Gaussian Discriminant Analysis
 - Generative & Discriminant comparison
 - Naive Bayes

A discriminative algorithm:

Learn $P(y|x)$

or learn $h_\theta(x) = \begin{cases} 0 \\ 1 \end{cases}$ directly

Generative learning algorithm

learns

$P(x|y)$

features

class

$P(y)$

class prior

so by Baye's rule

$$P(y=1|x) = \frac{P(x|y=1) P(y=1)}{P(x)}$$

$$P(x) = P(x|y=1) P(y=1) + P(x|y=0) P(y=0)$$

Gaussian Discriminant Analysis (GDA)

Suppose $x \in \mathbb{R}^n$ (drop $x_0=1$ convention)

Assume $P(x|y)$ is Gaussian

Σ

$$Z \sim \mathcal{N}(\vec{\mu}, \Sigma) \quad Z \in \mathbb{R}^n$$

$\downarrow \quad \downarrow$
 $\mathbb{R}^n \quad \mathbb{R}^{n \times n}$

$$E(Z) = \mu$$

$$\text{cov}(Z) = E[(Z - \mu)(Z - \mu)^T]$$

$$= E_{ZZ^T} - (E_Z)(E_Z)^T$$

$$E(Z) = E_Z$$

$$P(Z) = \frac{1}{(2\pi)^{n/2} |\Sigma|^{1/2}} \exp\left(-\frac{1}{2} (x - \mu)^T \Sigma^{-1} (x - \mu)\right)$$

GDA model $\Rightarrow$

$$P(X|y=0) = \frac{1}{(2\pi)^{n/2} |\Sigma|^{1/2}} \exp\left(-\frac{1}{2} (x - \mu_0)^T \Sigma^{-1} (x - \mu_0)\right)$$

Same Σ

$$P(X|y=1) = \frac{1}{(2\pi)^{n/2} |\Sigma|^{1/2}} \exp\left(-\frac{1}{2} (x - \mu_1)^T \Sigma^{-1} (x - \mu_1)\right)$$

$$P(y) = \phi^y (1 - \phi)^{1-y} \quad \boxed{P(y=0) = \phi}$$

Parameters: $\mu_0, \mu_1, \Sigma, \phi$

$\downarrow \quad \downarrow \quad \downarrow$
 $\mathbb{R}^n \quad \mathbb{R}^{m \times n} \quad \mathbb{R}$

Training set $\{x^{(i)}, y^{(i)}\}_{i=1}^m$

Joint likelihood

$$L(\phi, \mu_0, \mu_1, \Sigma) = \prod_{i=1}^m P(x^{(i)}, y^{(i)} | \phi, \mu_0, \mu_1, \Sigma)$$

$$L(\theta) = \prod_{i=1}^m P(x^{(i)} | y^{(i)}) P(y^{(i)})$$

Dis For discriminative learning, there was a difference in the likelihood function.

The below is also sometimes known as conditional likelihood.

$$L(\theta) = \prod_{i=1}^m P(y^{(i)} | x^{(i)}, \theta)$$

If we apply maximum likelihood estimation,

$$\max_{\phi, \mu_0, \mu_1, \Sigma} L(\phi, \mu_0, \mu_1, \Sigma)$$

$$= \log ()$$

$$\phi = \frac{\sum_{i=1}^m y^{(i)}}{m}$$

$$= \sum_{i=1}^m \frac{L\{y^{(i)} = 1\}}{m}$$

↓ This is the result.

$$L\{true\} = 1$$

$$L\{false\} = 0$$

$$\mu_0 = \frac{\sum_{i=1}^m L\{y^{(i)} = 0\} x^{(i)}}{\sum_{i=1}^m L\{y^{(i)} = 0\}}$$

← sum of feature vectors for examples with $y=0$

← # examples with $y=0$

$$\mu_1 = \frac{\sum_{i=1}^m L\{y^{(i)} = 1\} x^{(i)}}{\sum_{i=1}^m L\{y^{(i)} = 1\}}$$

$$\sum = \frac{1}{m} \sum_{i=1}^m (x^{(i)} - \mu_{y^{(i)}}) (x^{(i)} - \mu_{y^{(i)}})^T$$

Prediction : $\min_z (z-5)^2 = 0$

$\arg \min_z (z-5)^2 = 5$

Prediction

$$\arg \max_y P(y|x) = \arg \max_y \frac{P(x|y) P(y)}{P(x)}$$

since $P(x)$ is a const.

$$\arg \max_y P(x|y) P(y)$$

GDA

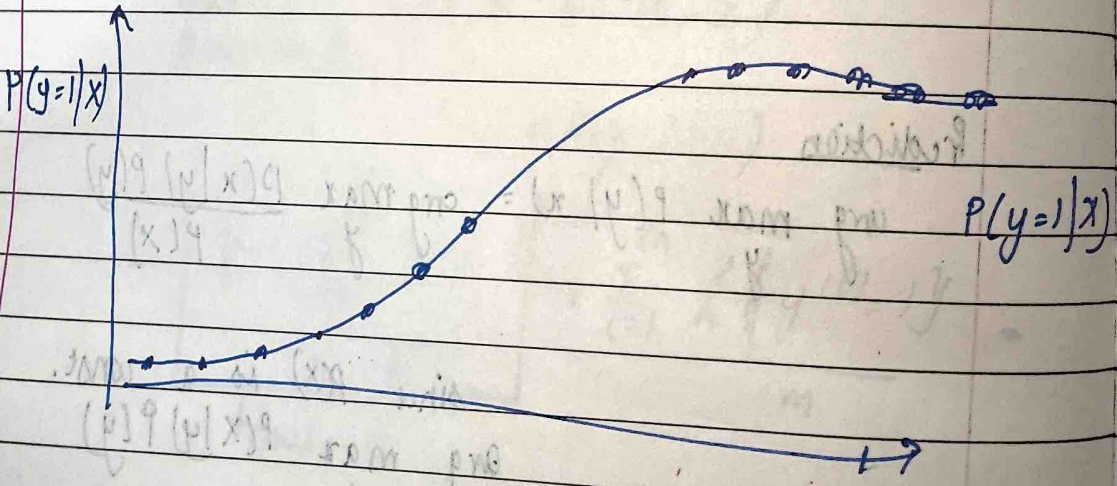
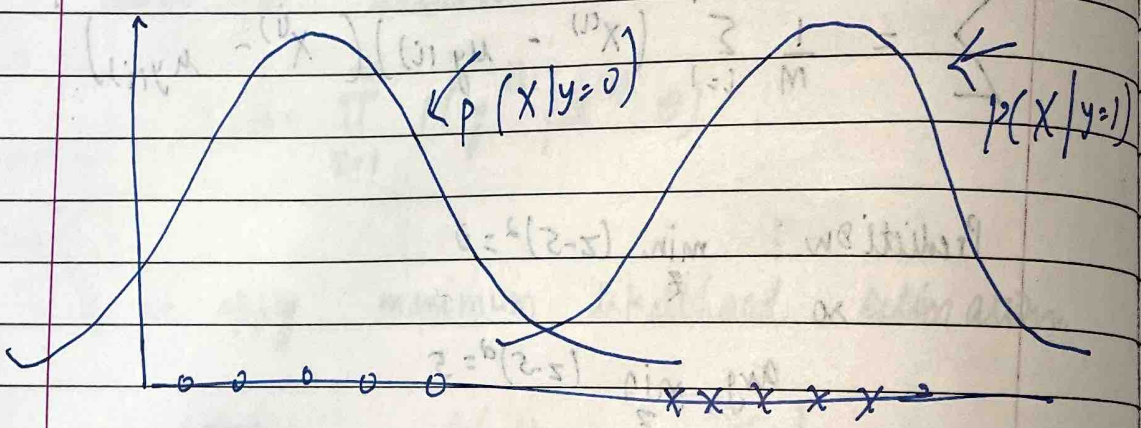
Lets compare to Logistic Regression.

For fixed $\phi, \mu_0, \mu_1, \Sigma$, lets

plot $P(y=1 | x, \phi, \mu_0, \mu_1, \Sigma)$ as a function of x .

by Bayes's Rule

$$\frac{P(x | y=1; \mu, \Sigma, \phi, \dots) P(y=1; \phi)}{P(x, \phi, \mu_0, \mu_1, \Sigma)}$$



generative approach
 GDA assumes

Discriminative
 Logistic
 regression

$$\left. \begin{aligned} X|y=0 &\sim \mathcal{N}(\mu_0, \Sigma) \\ X|y=1 &\sim \mathcal{N}(\mu_1, \Sigma) \\ y &\sim \text{Ber}(\phi) \end{aligned} \right\} \Rightarrow P(y=1|X) = \frac{1}{1+e^{-\phi^T X}}$$

"X₀ = 1"

Logistic
 But opposite

is not true.
 which means GDA
 takes stronger assumptions than Logistic Regression

$$\left. \begin{aligned} X|y=1 &\sim \text{Poisson}(\lambda_1) \\ X|y=0 &\sim \text{Poisson}(\lambda_0) \\ y &\sim \text{Ber}(\phi) \end{aligned} \right\} \Rightarrow \text{This also follows logistic regression}$$

Naive Bayes

Feature vector X

$$X = \begin{bmatrix} 1 \\ 0 \\ 0 \\ 0 \\ 1 \\ \vdots \\ \vdots \\ \vdots \end{bmatrix}$$

a
 aadwork
 aad wolf
 buy
 !

Zymurgy

$$x \in \{0, 1\}^n$$

$X_i = [\text{word } i \text{ appears in e-mail}]$

want to model $P(x|y), P(y)$

$2^{10,000}$ possible values of x .

So. In Naive Bayes

We assume x_i 's are conditionally independent given y .

$$P(x_1, \dots, x_{10000} | y) = P(x_1 | y) \cdot P(x_2 | y) \cdot P(x_3 | y) \cdot \dots \cdot P(x_{10000} | y)$$

$$= \prod_{i=1}^n P(x_i | y)$$

Parameters:

$$\phi_{j|y=1} = P(x_j=1 | y=1) \quad [\text{email is spam}]$$

$$\phi_{j|y=0} = P(x_j=1 | y=0)$$

$$\phi_y = P(y=1)$$

Joint Likelihood:

$$L(\phi_y, \phi_{j|y}) = \prod_{i=1}^m P(x^{(i)}, y^{(i)} | \phi_y, \phi_{j|y})$$

By applying MLE:

$$\phi_y = \frac{\sum_{i=1}^m \mathbb{1}\{y^{(i)} = 1\}}{m}$$

$$\phi_{j|y=1} = \frac{\sum_{i=1}^m \mathbb{1}\{x_j^{(i)} = 1 \wedge y^{(i)} = 1\}}{\sum_{i=1}^m \mathbb{1}\{y^{(i)} = 1\}}$$

$$\phi_{j|y=0} \rightarrow \frac{\sum_{i=1}^m \mathbb{1}\{x_j^{(i)} = 1 \wedge y^{(i)} = 0\}}{\sum_{i=1}^m \mathbb{1}\{y^{(i)} = 0\}}$$